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b1) Chéo hoá trực giao

Bài 1

$$A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$$

$$A.A^T = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix} \qquad A^T A = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}$$

Chéo hoá trực giao $AA^T$

$\det(AA^T - \lambda I) = 0$

$$\Leftrightarrow \begin{vmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{vmatrix} = 0$$

$$\Leftrightarrow \left[\begin{array}{l} \lambda_1 = 25 \\ \lambda_2 = 9 \end{array}\right.$$

$* \ \lambda_1 = 25$

$$\Rightarrow \begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow \left(\begin{array}{cc|c} -8 & 8 & 0 \\ 0 & 0 & 0 \end{array}\right) \Leftrightarrow \begin{cases} x_1 = t \\ x_2 = t \end{cases} \rightarrow x = (1, 1)t$$

$$\rightarrow u_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T \qquad (1)$$

Thứ ngày

$\lambda_2 = 9$

$\rightarrow \left(\begin{array}{ccc|c} 4 & 12 & 2 & 0 \\ 12 & 4 & -2 & 0 \\ 2 & -2 & -1 & 0 \end{array}\right)$

$\rightarrow \left(\begin{array}{ccc|c} 4 & 12 & 2 & 0 \\ 0 & +32 & 8 & 0 \\ 0 & 16 & 4 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 4 & 12 & 2 & 0 \\ 0 & 32 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$

$\rightarrow \begin{cases} x_1 = \frac{1}{4}.t \\ x_2 = \frac{-1}{4} t \\ x_3 = t \end{cases} \rightarrow x = t(1, -1, 4)$

$\rightarrow z = \left(\frac{+1}{3\sqrt{2}}, \frac{-1}{3\sqrt{2}}, \frac{4}{3\sqrt{2}}\right)^T$

$\lambda_3 = 0$

$\rightarrow \left(\begin{array}{ccc|c} 13 & 12 & 2 & 0 \\ 12 & 13 & -2 & 0 \\ 2 & -2 & 8 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 13 & 12 & 2 & 0 \\ 0 & 25 & -50 & 0 \\ 0 & 50 & -100 & 0 \end{array}\right)$

$\rightarrow \begin{cases} x_1 = -2t \\ x_2 = 2t \\ x_3 = 1.t \end{cases} \rightarrow x = (-2, 2, 1)t$

$\rightarrow v_3 = \left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}\right)^T$

$$\rightarrow V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{-2}{3} \\ \frac{1}{\sqrt{2}} & \frac{-1}{3\sqrt{2}} & \frac{2}{3} \\ 0 & \frac{4}{3\sqrt{2}} & \frac{1}{3} \end{bmatrix}$$

$$\Sigma = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}; \quad U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Bài 2. Kiểm tra tính lồi lõm.

$$f(x) = x_1^2 + 4x_1x_2 + 4x_2^2 - \ln(x_1^2 + x_2^2)$$

$$f'_{x_1} = 2x_1 + 4x_2 - \frac{1}{x_1^2+x_2^2}\cdot 2x_1$$

$$f'_{x_2} = 4x_1 + 8x_2 - \frac{1}{x_1^2+x_2^2}\cdot 2x_2$$

$$f''_{x_1^2} = 2 - \left(\frac{2}{x_1^2+x_2^2} - \frac{2x_1\cdot 2x_1}{(x_1^2+x_2^2)^2}\right)$$

$$= 2 - \cancel{\frac{2x_1^2-6x_1^2}{(x_1^2+x_2^2)^2}} - \frac{2x_2^2-2x_1^2}{(x_1+x_2)^2}$$

$$f''_{x_1x_2} = 4 - \left(\frac{2x_1\cdot 2x_2}{(x_1^2+x_2^2)^2}\right) = 4 - \frac{4x_1x_2}{(x_1^2+x_2^2)^2}$$

$$f''_{x_2x_2} = 8 - \left(\frac{2}{x_1^2+x_2^2} + \frac{-2x_2\cdot 2x_2}{(x_1^2+x_2^2)^2}\right)$$

$$= 8 - \frac{2x_1^2-2x_2^2}{(x_1^2+x_2^2)^2}$$

$$\Rightarrow \nabla^2 f(x) = \begin{pmatrix} 2 - \frac{2x_2^2-2x_1^2}{(x_1^2+x_2^2)^2} & 4 - \frac{4x_1x_2}{(x_1^2+x_2^2)^2} \\ 4 - \frac{4x_1x_2}{(x_1^2+x_2^2)^2} & 8 - \frac{2x_1^2-2x_2^2}{(x_1^2+x_2^2)^2} \end{pmatrix}$$

$$A_1 = 2 - \frac{2x_2^2-2x_1^2}{(x_1^2+x_2^2)^2} = \frac{2(x_1^4+2x_1^2x_2^2+x_2^4) - 2x_2^2 + 2x_1^2}{(x_1^2+x_2^2)^2}$$

$$A_1 = \frac{2x_1^4 + 4x_1^2x_2^2 + 2x_2^4 - 2x_2^2 + 2x_1^2}{(x_1^2+x_2^2)^2}$$

Xét $x_1 = 0$, $x_2 = 0.5 \Rightarrow \det(A_1) = -6 < 0$.

$\Rightarrow$ Hàm số không lồi, không lõm

$\Rightarrow$ Hàm số 0 lồi, 0 lõm

Câu 3.

| $x_1$ | $x_2$ | $y$ | $f_1(x)$ | $f_2(x)$ | $f_3(x)$ |
|---|---|---|---|---|---|
| 3 | 3 | 115 | 3 | 3 | 9 |
| 3.5 | 3 | 117 | 3.5 | 3 | 10.5 |
| 3 | 4 | 120 | 3 | 4 | 12 |
| 4 | 3.5 | 125 | 4 | 3.5 | 14 |
| 4 | 4 | 130 | 4 | 4 | 16 |

$f_1(x) = x_1,\quad f_2(x) = x_2,\quad f_3(x) = x_1 x_2$

$$A = \begin{bmatrix} 3 & 3 & 9 \\ 3.5 & 3 & 10.5 \\ 3 & 4 & 12 \\ 4 & 3.5 & 14 \\ 4 & 4 & 16 \end{bmatrix}$$

$$A^T A = \begin{pmatrix} 62.25 & 61.5 & 219.75 \\ 61.5 & 62.25 & 219.5 \\ 219.75 & 219.5 & 787.25 \end{pmatrix}$$

$$A^T y = \begin{bmatrix} 3 & 3.5 & 3 & 4 & 4 \\ 3 & 3 & 4 & 3.5 & 4 \\ 9 & 10.5 & 12 & 14 & 16 \end{bmatrix} \begin{bmatrix} 115 \\ 117 \\ 120 \\ 125 \\ 130 \end{bmatrix} = \begin{bmatrix} 2134.5 \\ 2133.5 \\ 7533.5 \end{bmatrix}$$

Thứ ngày

$$\rightarrow (A^T A)^{-1} \cdot A^T y \approx \begin{bmatrix} 28.39 \\ 25.26 \\ -5.34 \end{bmatrix}$$

$$\Rightarrow y = 28.39\, x_1 + 25.26\, x_2 - 5.4\, x_1 x_2.$$

$$y(x_1 = 4{,}5;\ x_2 = 4) = 131.595$$